1. 1/1

1. Find the integral of the 4^x

a. $\frac{4^x}{\log 4}+C$

b. $\frac{3^x}{\log 3}+C$

c. $\frac{2^x}{\log 2}+C$

d. $\frac{1^x}{\log x}+C$

e. $\frac{4}{\log 4}+C$

- (a)
- (b)
- (c)
- (d)
- (e)

2. 0/1

Evaluate $\int(2x+3)^2\,dx$

1. $1/6(2x+3)^2 + C$
2. $1/3(2x+3)^3 + C$
3. $5/6(2x+3)^3 + C$
4. $1/6(2x+3)^3 + C$
5. $1/6(3x+2)^3 + C$

- (a)
- (b)
- (c)
- (d)
- (e)

Correct answer

- (d)

3. 1/1

Evaluate $\int e^{-3x-3} dx$

1. $\frac{1}{4} e^{-3x-3} + C$

2. $\frac{1}{5} e^{-3x-3} + C$

3. $-\frac{1}{3} e^{-3x-3} + C$

4. $-\frac{1}{3} e^{-2x-3} + C$

5. $\frac{1}{2} e^{-3x-3} + C$

- (a)
- (b)
- (c)
- (d)
- (e)

4. 1/1

Evaluate $\int \frac{2x}{x^2+1}dx$

1. $\log(x^2+1) + C$
2. $\log x^2 + C$
3. $\log x + C$
4. $-\log|x^2+1| + C$
5. $\log|x^3+1| + C$

- (a)
- (b)
- (c)
- (d)
- (e)

5.

Evaluate $\int \frac{e^{x}+e^{-x}}{e^{x}-e^{-x}}\,dx$

1. $-\log|e^{x}-e^{-x}|+C$
2. $\log(e^{x}+e^{-x})+C$
3. $\log e^{x}+C$
4. $\log(-e^{-x})+C$
5. $\log|e^{x}-e^{-x}|+C$

- (a)
- (b)
- (c)
- (d)
- (e)

✓ 6. 1/1

Evaluate $\int x^2 e^{3x}\, dx$

1. $\frac{e^{3x}}{4}[x^2 - \frac{2x}{5} + \frac{2}{9}] + C$

2. $\frac{e^{3x}}{3}[x^2 - \frac{2x}{3} + \frac{2}{9}] + C$

3. $= \frac{e^{3x}}{3}[x^2 - \frac{2x}{3} + \frac{2}{9}] + C$

4. $= \frac{e^{3x}}{3}[x^2 - \frac{2x}{3}] + C$

5. $= [x^2 - \frac{2x}{3} + \frac{2}{9}] + C$

- ○ (a)
- ◉ (b) ✓
- ○ (c)
- ○ (d)
- ○ (e)

7.

1/1

Evaluate $\int \frac{2x-3}{x^2-3x+2}\,dx$

1. $\ln(x^2 - 3x) + C$

2. $\ln(3x + 2) + C$

3. $\ln(x^2 + 2) + C$

4. $\ln(x^2 - 3x + 2) + C$

5. $\ln(x^2 + 3x + 2) + C$

(a)

(b)

(c)

(d)

(e)

8.

1/1

Which of the following define reduction formula for $\int x^n e^x\,dx$

1. $I_{n-1} = x^n e^x - nI_n$
2. $I_n = x^n e^x - nI_{n-2}$
3. $I_n = x^n e^x - nI_{n-1}$
4. $I_n = nI_{n-1} - x^n e^x$
5. $I_{n-2} = x^n e^x - nI_n$

(a)

(b)

(c)

(d)

(e)

9. 1/1

Compute $\int_0^2 (4x^2 - 5x + 7)dx$

1. $\frac{40}{3}$
2. $\frac{44}{5}$
3. $\frac{42}{3}$
4. $\frac{43}{3}$
5. $\frac{44}{3}$

- (a)
- (b)
- (c)
- (d)
- (e)

10.

1/1

Analyze $\int_2^3 \frac{x}{1+x^2} dx$

a. $\frac{1}{2}\log 2$

b. $\log 2$

c. $\frac{1}{3}\log 2$

d. $\frac{3}{2}\log 2$

e. $\frac{1}{2}\log 3$

(a)

(b)

(c)

(d)

(e)

✓ 11. 1/1

Find the area bounded by the curve $x = y$, y-axis and the lines $y = 0$, $y = 3$.

a. $\frac{9}{2}$ square unit

b. $\frac{9}{5}$ square unit

c. $\frac{7}{2}$ square unit

d. $\frac{5}{2}$ square unit

e. $-\frac{9}{2}$ square unit

- (a) ✓
- (b)
- (c)
- (d)
- (e)

12. 1/1

If $\frac{dy}{dx} = (2x+1)^2$ $find\ y.$

a. $\frac{4x^3}{3} + 2x^2 + x + C$

b. $\frac{4x^3}{3} - 2x^2 - x + C$

c. $\frac{4x^3}{3} + 2x^2 - x - C$

d. $\frac{4x^3}{3} + 2x^2 - x + C$

e. $-\frac{4x^3}{3} + 2x^2 + x + C$

(a)

(b)

(c)

(d)

(e)

13.

1/1

The gradient of a curve is 4x + 6 and it passes through the point (1, 6),

find the equation.

1. $Y = 2x^2 - 6x - 2$
2. $Y = 2x^2 + 6x + 2$
3. $Y = 2x^2 + 6x - 2$
4. $Y = -2x^2 + 6x - 2$
5. $Y = 3x^2 - 6x - 2$

(a)

(b)

(c)

(d)

(e)